ខាងក្រោមនេះគឺជា Active–Active + Weight Logic (Production-ready) សម្រាប់ Local Authoritative DNS (BIND9)

👉 មាន Health Check + Weight-based Load Balancing + Auto Failover

1 Concept: Active–Active + Weight

🎯 គោលគំនិត

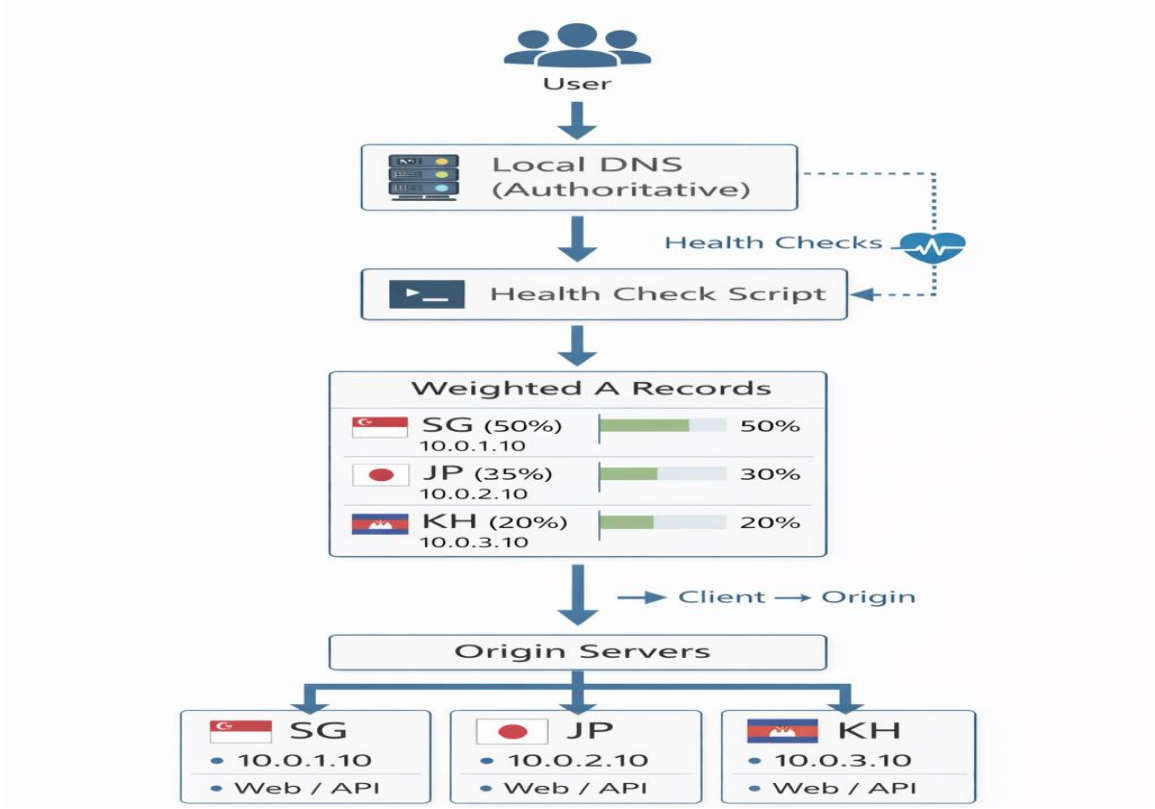
- Server ច្រើនដំណើរការពេលតែមួយ (Active–Active)
- DNS return IP តាម Weight
- Server down → Auto remove
- Server ឡើងវិញ → Auto add

Example Weight

Region	IP	Weight
SG	10.0.1.10	50
JP	10.0.2.10	30
KH	10.0.3.10	20

➡️ DNS round-robin តាម ratio 5:3:2

2 Architecture Flow



3 File Locations (Standard)

Item	Path
Zone file	/etc/bind/zones/db.example.com
Script	/usr/local/bin/dns-weighted-health.sh
Log	/var/log/dns-weighted-health.log
Temp	/tmp/dns_weighted.tmp

4 Zone File (Base – Static Part)

📍 /etc/bind/zones/db.example.com

```
$TTL 60
@   IN  SOA ns1.example.com. admin.example.com. (
        2026010101
        3600
        900
        604800
        60  )
    IN  NS  ns1.example.com.
ns1 IN  A   192.168.1.10
; Dynamic A records below
```

5 Active–Active + Weight Script

📍 /usr/local/bin/dns-weighted-health.sh

```
#!/bin/bash
ZONE="example.com"
ZONE_FILE="/etc/bind/zones/db.example.com"
LOG="/var/log/dns-weighted-health.log"
TMP="/tmp/dns_weighted.tmp"
HEALTH_PATH="/health"
TIMEOUT=3
# IP + WEIGHT
SG_IP="10.0.1.10"; SG_W=5
JP_IP="10.0.2.10"; JP_W=3
KH_IP="10.0.3.10"; KH_W=2
```

```
check( ) {
  curl -fs --max-time $TIMEOUT http://$1$HEALTH_PATH >/dev/null
}
log( ) {
  echo "$( date '+%F %T' ) $1" >> $LOG
}
> $TMP
ACTIVE=0
add_weighted( ) {
  IP=$1
  W=$2
  for i in $( seq 1 $W ); do
    echo "@ IN A $IP" >> $TMP
  done
}
log "Health check started"
if check $SG_IP; then
  add_weighted $SG_IP $SG_W
  ACTIVE=1
  log "SG UP ( weight $SG_W )"
else
  log "SG DOWN"
fi
if check $JP_IP; then
  add_weighted $JP_IP $JP_W
  ACTIVE=1
  log "JP UP ( weight $JP_W )"
else
  log "JP DOWN"
fi
if check $KH_IP; then
  add_weighted $KH_IP $KH_W
```

```
  ACTIVE=1
  log "KH UP (weight $KH_W)"
else
  log "KH DOWN"
fi
if [ $ACTIVE -eq 0 ]; then
  log "CRITICAL: ALL ORIGINS DOWN"
  exit 1
fi
# Update zone
sed -i '/^@ IN A/d' $ZONE_FILE
cat $TMP >> $ZONE_FILE
# Update serial
SERIAL=$(date +%Y%m%d%H)
sed -i "s/^[ \t]*[0-9]\{10\}/        $SERIAL/" $ZONE_FILE
rndc reload $ZONE
log "Zone reloaded successfully"
```

6 Permissions

```
chmod +x /usr/local/bin/dns-weighted-health.sh
chown root:bind /etc/bind/zones/db.example.com
chmod 640 /etc/bind/zones/db.example.com
```

7 Cron (Run every 30s)

📍 /etc/cron.d/dns-weighted-health

```
* * * * * root /usr/local/bin/dns-weighted-health.sh
* * * * * root sleep 30 && /usr/local/bin/dns-weighted-health.sh
```

8 How Weight Works (Important)

Generated A records

```
SG SG SG SG SG
JP JP JP
KH KH
```

➡️ Resolver randomizes order

➡️ Probability ≈ Weight %

9 Failover Scenarios

Scenario	DNS Result
All UP	SG 50% / JP 30% / KH 20%
SG down	JP 60% / KH 40%
SG + JP down	KH 100%
One back	Auto restored

10 Testing

```
watch -n 1 "dig +short example.com @192.168.1.10"
```

Simulate failure:

```
iptables -A INPUT -p tcp --dport 80 -j DROP
```

11 Best Practices ✅

- ✔ TTL ≤ 60s
- ✔ Lightweight /health
- ✔ Separate Secondary DNS
- ✔ Monitor log & alert
- ✔ Version-control script

12 Limitations ❌

- DNS-based（not session-aware）
- No Geo routing
- Cache dependent

👉 For global scale → Cloudflare / Route53 / PowerDNS

✅ Summary

Active–Active + Weight DNS LB gives you:

- Simple
- Fast
- Cost-effective
- Automatic failover